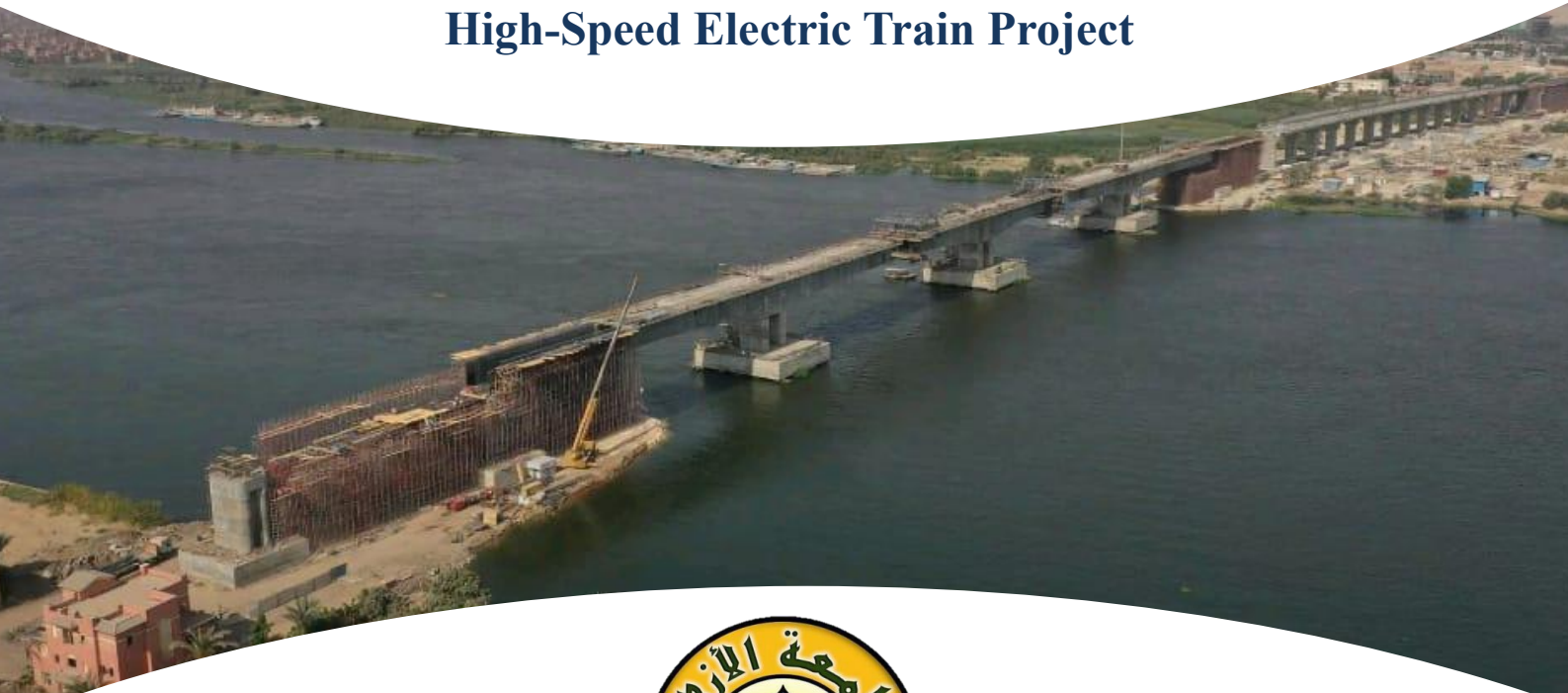




El Soaada Group

Nile Bridge – Marazik

High-Speed Electric Train Project



Summer internship

REPORT

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3rd-Construction Department



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Introduction

At The High-Speed Electric Train Project

During the internship, we learn and saw the implementation of:

Substructure Works

- Pile execution using bentonite slurry (soft soil near the Nile).
- Reinforcement cage preparation for piles.
- Concreting process and concrete delivery line setup.

Foundation & Piers

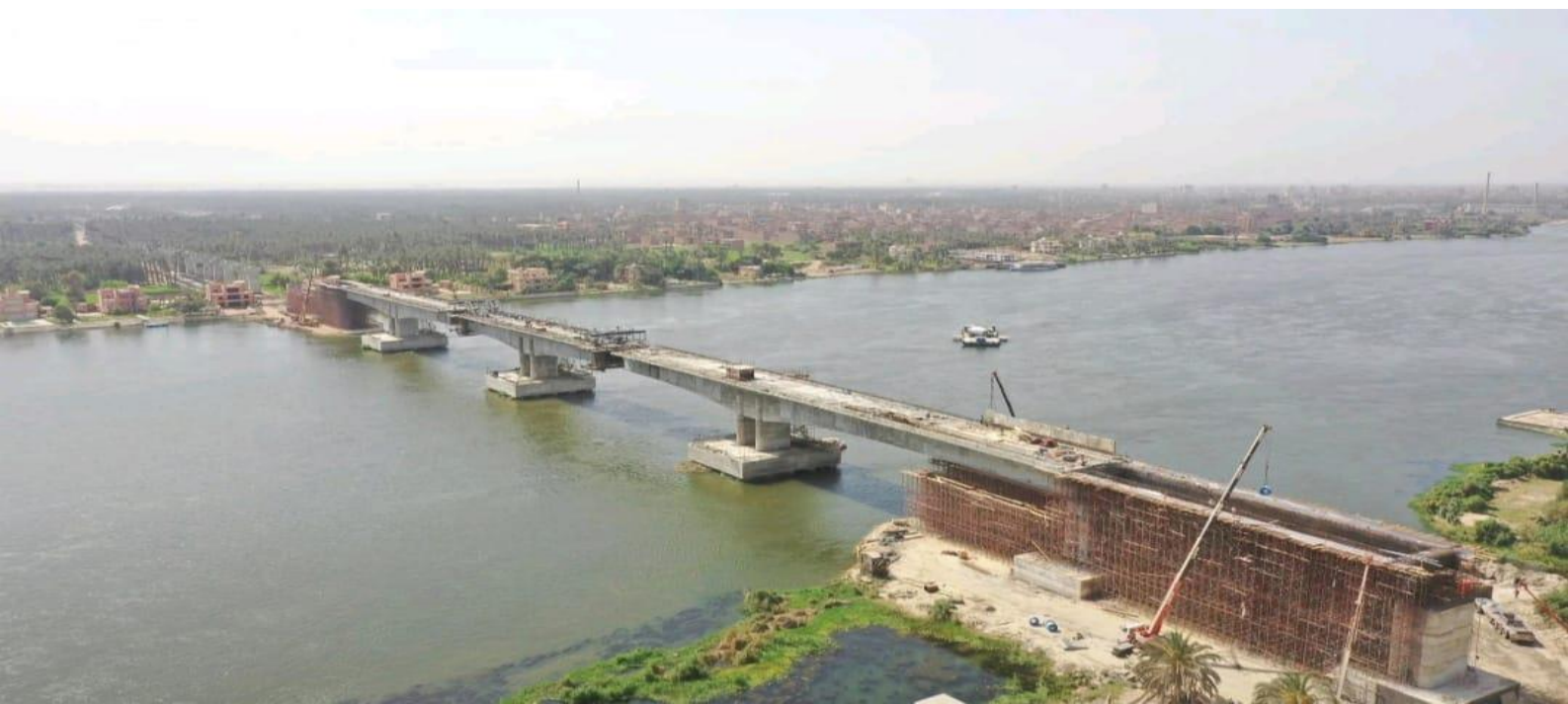
- Large dimensions → Metal formwork used for mass concrete.
- Construction of piers.
- Introduction to bearings between substructure and superstructure.

Superstructure Works

- Types used: Omega Girders, Box Sections, Cantilever Carriage (Nile crossing).
- Reinforcement details of Omega girders.
- Post-tensioning process.
- Omega girder formwork (mold) preparation.
- Cantilever Carriage implementation

Batching Plant Visit

- Components of a ready-mix concrete plant.
- Slump test for workability.
- Casting concrete cubes for strength testing.



Project Overview

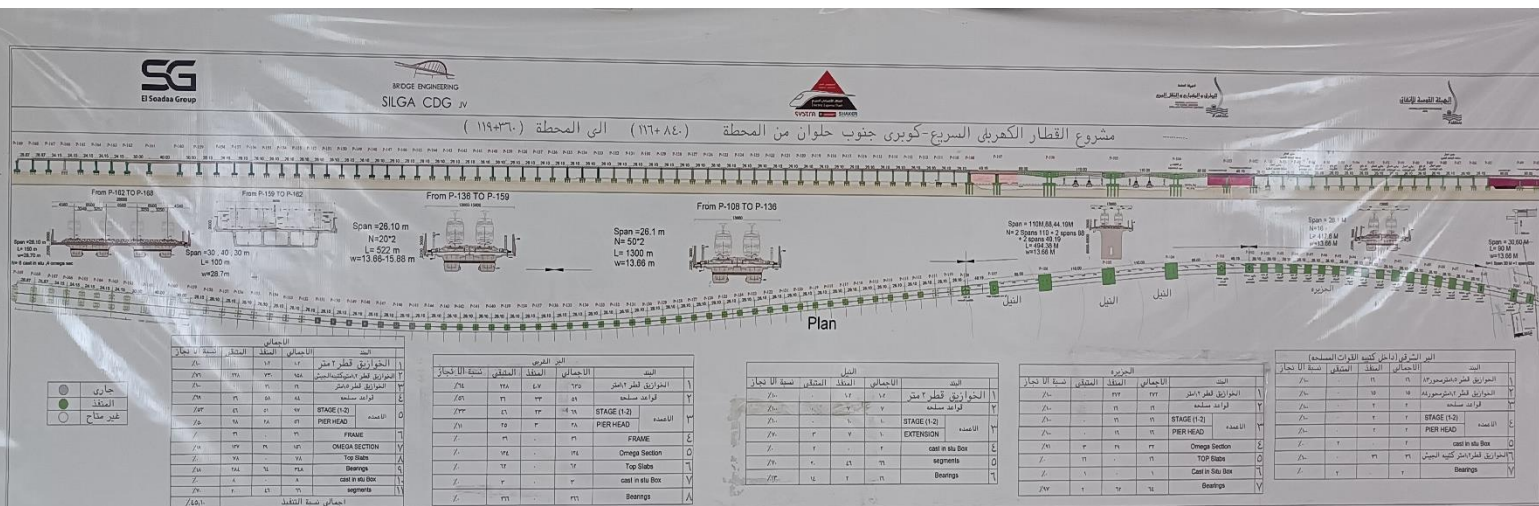
The High-Speed Electric Train Project in Egypt is one of the country's most ambitious infrastructure developments, aiming to provide fast, safe, and environmentally friendly transportation.

Implemented by an international consortium led by Siemens, in cooperation with hundreds of Egyptian national companies. The total number of participating entities, including construction companies, engineering consultancy offices, and suppliers, is estimated at around 3,000. Among these, approximately 200 companies are directly involved in on-site construction works, with 23 companies specifically responsible for station construction.

the network consists of three main lines:

- The first line, extending approximately 660 km, connects Ain Sokhna on the Red Sea to Alamein and Matrouh on the Mediterranean coast.
- The second line will link 6th of October City to Aswan over a length of about 1,100 km.
- The third line will connect Hurghada and Safaga to Qena and Luxor, spanning around 225 km.

The trains are designed to operate at speeds of up to **250 km/h** for passengers and approximately 120 km/h for freight, powered entirely by electricity to minimize environmental impact.



Sup structure

Initial Preparations

- Topographic survey for the designated site.
- Obtaining Site Handover and official Construction Permit from the authorities.
- Setting out works and site demarcation before starting piling activities.



Piles

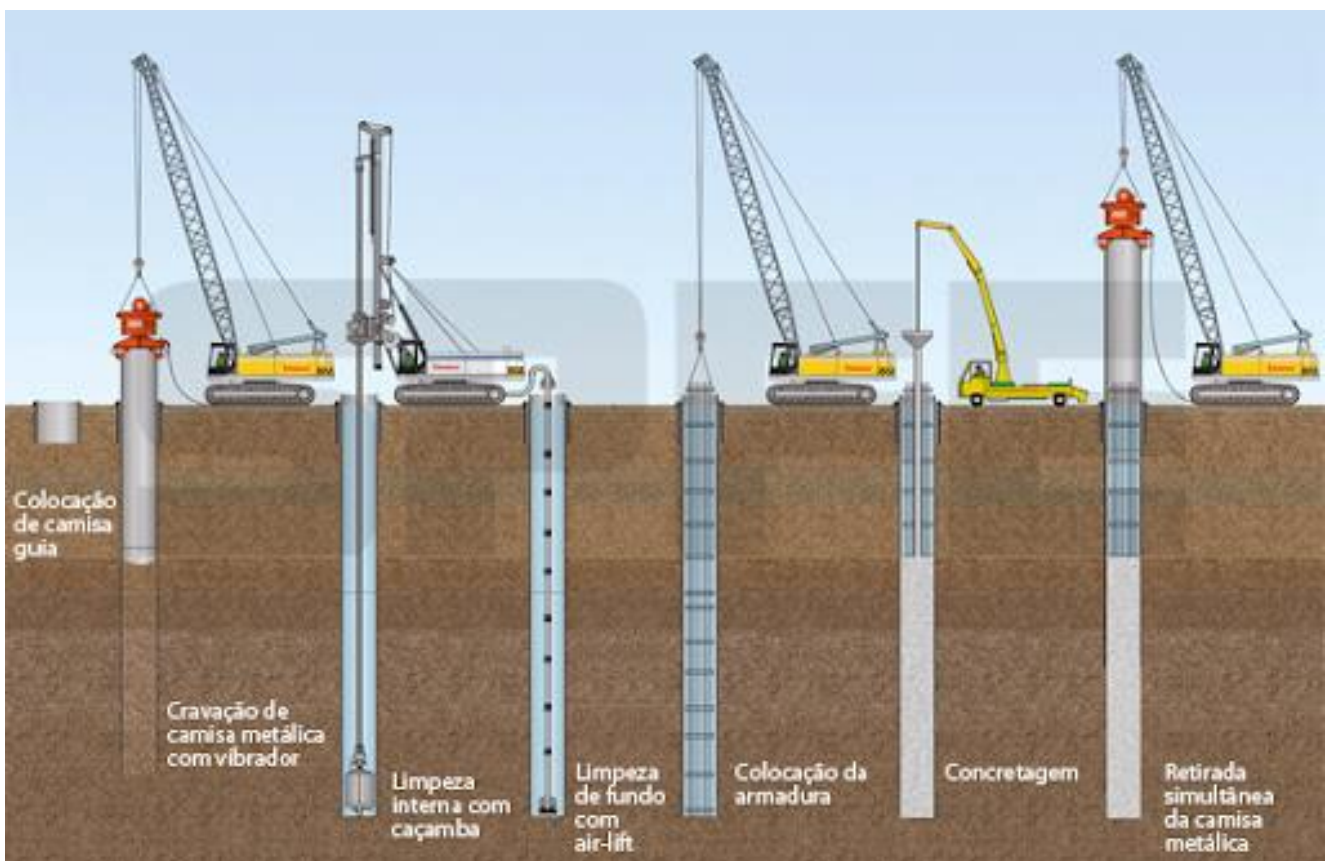


Cast-in-Place Concrete Piles

- Execution: Drilled on site, reinforcement placed, concrete poured. Quality: Depends on site/workmanship.
- Flexibility: Adjustable length & diameter.
- Speed: Slower (drilling + pouring).
- Cost: Lower transport cost, more labour time.
- Impact: Minimal vibration & noise.
- Length Limit: 40–60 m+ (depends on soil & equipment).

Precast Concrete Piles

- Execution: Made in the factory, transported, and driven on site.
- Quality: High (controlled production).
- Flexibility: Fixed size — hard to modify.
- Speed: Fast installation.
- Cost: Higher (transport + heavy equipment).
- Impact: Vibration & noise during driving.
- Length Limit: 25–30m (transport/handling limit).



Sup structure

Cast-in-Place Concrete Piles with Bentonite

Purpose of Bentonite

- Slurry used to support borehole walls during drilling.
- Prevents collapse in loose or water-bearing soils.

Execution Steps

- Drilling the borehole to the required depth.
- Filling with bentonite slurry to stabilize the walls.
- Placing a reinforcement cage inside the borehole.
- Pouring concrete using a tremie pipe (keeps bentonite out).
- Displacing bentonite upwards as concrete fills the hole.

Advantages

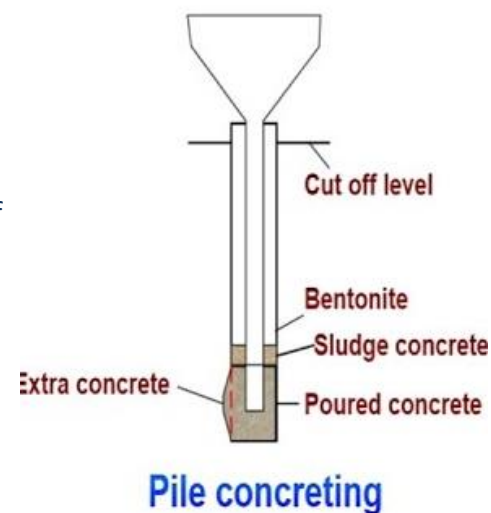
- Can be used in soft or saturated soils.
- Maintains borehole shape until concreting.
- Allows greater depths (up to 60 m+).

Limitations

- Requires careful bentonite quality control.
- Disposal of used slurry adds cost & environmental work.

Bentonite quality control

Test	Method / Tool	Acceptable Range	Purpose
Density	Mud Balance	1.03 – 1.12 g/cm ³	Stability & support of borehole
Viscosity	Marsh Funnel (sec/litre)	32 – 50 sec	Flow & coating of walls
Sand Content	Sand Content Kit	< 4%	Avoid weakening slurry
pH	pH Meter / Paper	8 – 11	Optimal gel formation
Filtration Loss	Filter Press	Low (as per project spec)	Prevent water seepage



Sup structure

Pile Cap

- Definition: A Thick reinforced concrete slab that connects the tops of piles and transfers the load from the column or pier to the piles.
- Function: Distributes structural loads evenly to all piles. Provides stability to the foundation system.
- Construction Steps:
 1. Excavation to pile cap level.
 2. trim pilesHead to design level & clean surface.
 3. set shuttering formwork for the required dimensions.
 4. top & bottom layers Reinforcement tied according to drawings.
 5. Pouring mass concrete to reduce shrinkage cracks.
 6. Keep the surface moist to ensure strength.



Pier column

- Definition: Vertical reinforced concrete member that transfers loads from the bridge superstructure to the pile cap or directly to the foundation.
- Function:
 - Supports the girders/deck of the bridge.
 - Transfers vertical loads (traffic, self-weight) and horizontal loads (wind, water current, braking).
 - Provides stability against bending and shear forces.
- Construction Steps:
 1. Pile Cap Completion — pier sits on top.
 2. Formwork Erection — usually circular or rectangular steel/ply shuttering.
 3. Reinforcement Placement — longitudinal bars + ties/stirrups.
 4. Concrete Pouring — often in lifts (stages) for tall piers.
 5. Curing — maintain moisture for strength gain.



Superstructure

Omega Girders

- Type: Precast prestressed girder (cross-section like Ω).
- Function: Supports bridge deck slab & transfers loads to piers/abutments.
- Advantages: Lightweight, strong, easy to install, aesthetic profile.
- Length Limit: ~30–40 m.



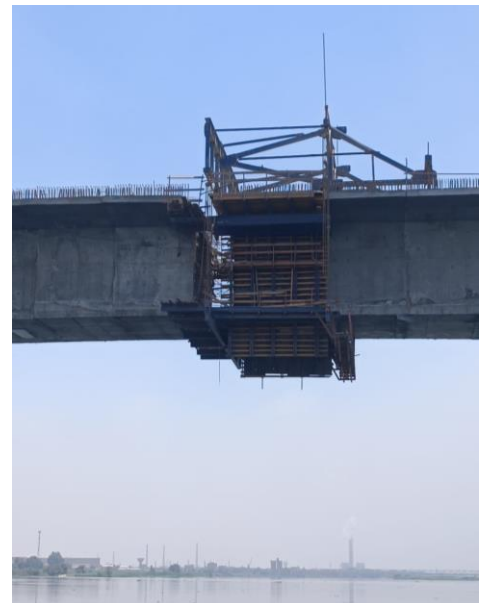
Box Section Girder

- Type: Hollow rectangular or trapezoidal cross-section (closed box).
- Function: High torsional stiffness — ideal for curved bridges or wide decks.
- Advantages:
 - Very strong in bending & torsion.
 - Can carry heavy loads over long spans.
 - Aerodynamic shape reduces wind effects.
- Length Limit: Can exceed 50 m (depending on design).



Cantilever Carriage

- Type: Special movable steel frame/platform.
- Function: Supports formwork & workers for in-situ casting of bridge segments (especially for the cantilever method).
- Advantages:
 - Allows casting without scaffolding from ground/water.
 - Moves along the span as work progresses.
- Use: Common in segmental and cable-stayed bridges.



Superstructure

Bearings (Elastomeric / Pot Bearings)

- Definition: Support elements placed between the superstructure and piers.
- Function: Transfer loads while allowing movements due to temperature and loads.
- Key Point: Bearing type is selected based on load capacity and movement requirements.



Expansion Joints

- Definition: Flexible connections between bridge segments.
- Function: Accommodate expansion/contraction caused by temperature changes.
- Key Point: Regular maintenance is essential to avoid deterioration.



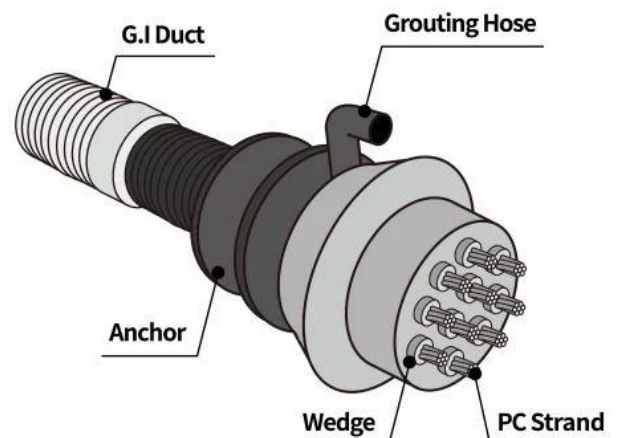
Precast Segments

- Type: Precast prestressed girder (cross-section like Ω).
- Function: Supports bridge deck slab & transfers loads to piers/abutments.
- Advantages: Lightweight, strong, easy to install, aesthetic profile.
- Length Limit: ~30–40 m.



Post-tensioning System

- Definition: Applying tension to steel tendons after concrete curing.
- Function: Increases bending capacity and allows for longer spans.
- Key Point: Demands accurate stress application and duct installation



Ready-Mix Concrete Batching Plant

We visit the batching plant that belongs to the project
So, to observe the process of producing ready-mix concrete for the project.

Main Components:

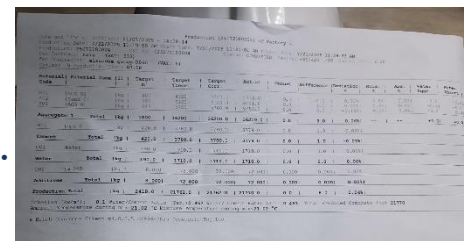
- Aggregate storage bins (sand, gravel, crushed stone),
- Cement silos
- Water tanks
- Weighing system
- Mixing unit (drum or twin-shaft mixer)
- Control room for automated operation.

Process Overview:

- Weighing of aggregates, cement, and water according to mix design.
- Mixing in the batching unit.
- Quality control tests (slump test, temperature, etc.).
- Loading into transit mixers for delivery.

Key Observations:

- High level of automation ensures accuracy.
- Quality tests are done on every batch before delivery..



Material	Batch No.	Weight (kg)	Volume (m³)	Water (kg)	Water (m³)	Water/Cement Ratio	Slump (cm)	Temperature (°C)	Remarks
Cement	1	3500	1.40	100	0.10	0.03	10	25	
Gravel	2	1200	0.72	0	0				
Sand	3	800	0.48	0	0				
Water	4	100	0.10	100	0.10	0.03			
Total		5600	2.60	200	0.20	0.07	10	25	



Concrete Quality Control Tests

Slump Test

- Measures workability of fresh concrete.
- Ensures it meets the required slump range for the project.



Temperature Test

- Measure fresh concrete temperature on site.
- Ensure it's within limits to prevent setting & strength issues.



Cube Test:

- Cast concrete cubes and cure them.
- Test compressive strength at 7 & 28 days.

